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Results in Mathematics



Structural Properties, Complete Monotonicity, Bounds, and Analytic Approximations with Bounded Remainders for a Wallis-Type Gamma Ratio

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Abstract. In this paper, we investigate several completely monotonic functions involving the Psi function, which is defined as the logarithmic derivative of the Gamma function. By exploiting the analytic and monotonic properties of these functions, we develop three new approximation formulas for a Wallis-type ratio with bounded remainder terms. These approximations lead to a number of new inequalities that are better than some existing bounds in the literature. Finally, an open problem is proposed concerning the monotonic behavior of the remainder functions arising from the new approximations.

Mathematics Subject Classification. 33B15, 41A60, 26D15, 41A30.

Keywords. Wallis' ratio, Gamma function, Psi function, Asymptotic approximation, Speed of convergence.

1. Introduction

The ratio

$$W(x) = \frac{1}{\sqrt{\pi}} \frac{\Gamma(x+1/2)}{\Gamma(x+1)}, \qquad x > 0, \tag{1}$$

where $\Gamma(x)$ denotes the Euler Gamma function, arises in many areas of mathematics, including asymptotic analysis, special functions, probability, and combinatorics. This ratio generalizes the coefficients appearing in Wallis' classical product for π, and it frequently appears in integral representations and probability density functions involving normalizing constants.

A fundamental property of $W(x)$ is its asymptotic behavior as x grows. One of the earliest systematic methods for describing this behavior is derived from Stirling's asymptotic expansion for the gamma function [1],

$$\Gamma(x) \sim \sqrt{2\pi}\, x^{x-\frac{1}{2}} e^{-x} \left[1 + \frac{1}{12x} + \frac{1}{288x^2} + \dots\right], \qquad x \to \infty,$$

which leads to an asymptotic series for $W(x)$ in inverse powers of x:

$$W(x) \sim \frac{x^{-1/2}}{\sqrt{\pi}} \left[1 - \frac{1}{8x} + \frac{1}{128x^2} - \frac{5}{1024x^3} + \frac{21}{32768x^4} + \cdots\right], \quad x \to \infty. \tag{2}$$

This expansion provides increasingly accurate approximations for large x but can be inefficient for moderate values of the argument.

Over the last decades, numerous authors have investigated both asymptotic series and alternative approximation techniques for Gamma function ratios. In particular, Guo et al. establish sharp upper and lower bounds for approximating the Wallis' ratio, and derive the asymptotic expansion [2]

$$W(x) \sim \sqrt{\frac{e}{\pi}} \left(1 - \frac{1}{2x}\right)^x \frac{\sqrt{x-1}}{x}, \quad x \to \infty. \tag{3}$$

Qi and Mortici [3] establish for the Wallis ratio some best approximation formulas, inequalities and the asymptotic expansion

$$W(x) \sim \sqrt{\frac{e}{\pi}} \left(1 - \frac{1}{2x}\right)^x \frac{1}{\sqrt{x}} \exp\left(\frac{1}{24x^2} + \frac{1}{48x^3} + \frac{1}{160x^4} + \frac{1}{960x^5} + \cdots\right) \quad x \to \infty. \tag{4}$$

Guo et al. derive sharp upper and lower bounds for the Wallis ratio and obtain the asymptotic expansion [4]

$$W(x) \sim \sqrt{\frac{e}{\pi}} \left(1 - \frac{1}{2x}\right)^{x+1/2} \left(x - \frac{3}{2}\right)^{-1/2}, \quad x \to \infty. \tag{5}$$

Liu et al. present the following new formula [5]

$$W(x) \sim \sqrt{\frac{e}{\pi x}} \left(1 - \frac{1}{2x}\right)^x \exp\left(\frac{1/24}{x^2 - \frac{1}{2}x + 1}\right), \quad x \to \infty. \tag{6}$$

Yang and Tian establish the following new asymptotic expansion [6]

$$W(x) \sim \frac{1}{\sqrt{\pi(x+1/4)}} e^{-\frac{1}{64x^2} + \frac{1}{128x^3} - \frac{1}{2048x^4} + \cdots}, \quad x \to \infty.$$

Cristea deduce the following approximation [7]

$$W(x) \approx \frac{1}{\sqrt{\pi x}} e^{\frac{1}{2\sqrt{2}}\left[\tan^{-1}(2\sqrt{2}x) - \frac{\pi}{2}\right] - \frac{3}{2560\, x^5} + \frac{33}{28672\, x^7}}, \quad x \to \infty \tag{7}$$

motivated by the Mahmoud et al.'s inequality [8]

$$W(x) < \frac{1}{\sqrt{\pi x}} e^{\frac{1}{2\sqrt{2}}\left[\tan^{-1}(2\sqrt{2}x)-\frac{\pi}{2}\right]}, \quad x \geq 0. \tag{8}$$

Also, Cristea give the following improvement of inequality (8),

$$\begin{aligned} &\frac{1}{\sqrt{\pi x}} \exp\left[-\frac{1}{2\sqrt{2}} \cot^{-1}\left(2\sqrt{2}x\right) - \frac{3}{2560x^5}\right] \\ &< W(x) < \frac{1}{\sqrt{\pi x}} \exp\left[-\frac{1}{2\sqrt{2}} \cot^{-1}\left(2\sqrt{2}x\right) - \frac{3}{2560x^5} + \frac{33}{28672x^7}\right], \\ &\quad x > 0. \end{aligned} \tag{9}$$

Beyond several expansions, there is a well-developed literature on rigorous inequalities that bound $W(x)$. Such inequalities are useful not only for theoretical analysis but also for practical computations where exact Gamma evaluations are not be possible. For more details, we refer to [9–15] and the references therein.

In this paper, we study several completely monotonic functions involving the Psi function, $\psi(x)$, which is the logarithmic derivative of the Gamma function, i.e., $\psi(x) = \frac{d}{dx} \ln \Gamma(x) = \frac{\Gamma'(x)}{\Gamma(x)}$. We also derive three approximation formulas for the ratio $W(x)$ with bounded remainders. Consequently, we establish several inequalities for $W(x)$. Finally, we propose an open question concerning the monotonic behavior of the remainder functions appearing in our new approximations.

In this work, Mathematica (version 10) was employed to carry out all necessary symbolic and numerical computations.

2. Main Results

Lemma 2.1. *The function*

$$G_1(x) = \frac{-1600x^5 + 400x^4 - 960x^3 + 190x^2 - 144x + 31}{32x^2\left(10x^2+3\right)^2} + \psi(x+1) - \psi(x+1/2)$$

is completely monotonic for $x > 0$. As a consequence, the Wallis ratio satisfies the inequality

$$\frac{\Gamma\left(x+1/2\right)}{\Gamma(x+1)} > \frac{1}{\sqrt{x}}\, e^{-\frac{120x^2+31}{96x\left(10x^2+3\right)}}, \quad x > 0. \tag{10}$$

Proof. Consider the function

$$F_1(x) = -\ln\left(\frac{\sqrt{x}\Gamma\left(x+\frac{1}{2}\right)}{\Gamma(x+1)}\right) - \frac{120x^2+31}{96x\left(10x^2+3\right)}, \quad x > 0$$

and use the relation [1]

$$\psi(x) = \int_0^\infty \frac{e^{-t} - e^{-xt}}{1-e^{-t}} dt, \quad x > 0 \tag{11}$$

to get

$$\begin{aligned}\frac{d}{dx}F_1(x) &= G_1(x)\\ &= \int_0^\infty \frac{31t + e^{t/2}(31t-144) + 5\left(e^{t/2}+1\right)t\cos\left(\sqrt{\frac{3}{10}}t\right) + 144}{288\left(e^{t/2}+1\right)}\\ &\quad \times e^{-xt}dt.\end{aligned}$$

Using the inequality [16]

$$\cos(t) > -\frac{t^6}{720} + \frac{t^4}{24} - \frac{t^2}{2} + 1, \quad t > 0$$

we have

$$\begin{aligned}\cos(t) - \frac{12-5t^2}{t^2+12} &> -\frac{t^6}{720} + \frac{t^4}{24} - \frac{t^2}{2} + 1 - \frac{12-5t^2}{t^2+12}\\ &= \frac{t^6\left(18-t^2\right)}{720\left(t^2+12\right)} > 0,\ 0 < t < \sqrt{18} \approx 4.242\end{aligned}$$

and also

$$\cos(t) - \frac{12-5t^2}{t^2+12} > -1 - \frac{12-5t^2}{t^2+12} = \frac{4\left(t^2-6\right)}{t^2+12} > 0, \quad t > \sqrt{6} \approx 2.449.$$

Therefore we obtain

$$\cos(t) > \frac{12-5t^2}{t^2+12}, \quad t > 0. \tag{12}$$

Then

$$\begin{aligned}&31t + e^{t/2}(31t-144) + 5\left(e^{t/2}+1\right)t\cos\left(\sqrt{\frac{3}{10}}t\right) + 144\\ &> \frac{6}{t^2+40}\left[t(t(t+24)+240) + e^{t/2}(t((t-24)t+240)-960)+960\right]\\ &> \frac{6}{t^2+40}\sum_{n=7}^{\infty}\left\{2^{3-n}(n-6)(n-5)(n-4)\right\}\frac{t^n}{n!} > 0, \quad t > 0.\end{aligned}$$

Hence $G_1(x)$ is completely monotonic function and then $F_1(x)$ is increasing function for $x > 0$. Also, as $x \to \infty$, the logarithm of the Gamma function admits the Stirling-type asymptotic expansion [1]:

$$\ln\Gamma(x) \sim \left(x - \frac{1}{2}\right)\ln x - x + \frac{1}{2}\ln(2\pi) + \sum_{n=1}^{\infty}\frac{B_{2n}}{2n(2n-1)x^{2n-1}}, \tag{13}$$

where $B_n's$ are Bernoulli numbers, then we have

$$F_1(x) \sim -\frac{257}{358400x^7} + \frac{3551}{2304000x^9} + \ldots, \quad x \to \infty$$

and we get

$$\lim_{x\to\infty} F_1(x) = 0.$$

Hence the function $F_1(x)$ is negative function for $x > 0$, which completes the proof. □

Lemma 2.2. *The function*

$$F_2(x) = \ln\left(\frac{\sqrt{x}\Gamma\left(x+\frac{1}{2}\right)}{\Gamma(x+1)}\right) + \frac{3x}{24x^2+1}, \quad x > 0$$

is increasing for $x > 0$. As a consequence, the Wallis ratio satisfies the inequality

$$\frac{\Gamma\left(x+1/2\right)}{\Gamma(x+1)} < \frac{1}{\sqrt{x}}\, e^{-\frac{3x}{24x^2+1}}, \quad x > 0. \tag{14}$$

Proof. From

$$\frac{d}{dx}F_2(x) = \frac{3-72x^2}{\left(24x^2+1\right)^2} + \frac{1}{2x} + \psi\left(x+1/2\right) - \psi(x+1)$$

and using the relation [1]

$$\psi(x+1) = \psi(x) + \frac{1}{x}, \quad x > 0 \tag{15}$$

we have for $x > 0$, that

$$\begin{aligned}&\frac{d}{dx}\left\{F_2(x) - F_2(x+2)\right\}\\ &= \frac{2x(x+2)(48x(x+2)(1116x(x+2)+6269)+66145)+28227}{x(x+1)(x+2)(2x+1)(2x+3)\left(24x^2+1\right)^2\left(24x(x+4)+97\right)^2} > 0.\end{aligned}$$

Therefore $\frac{d}{dx}F_2(x) > \frac{d}{dx}F_2(x+2)$ for $x > 0$, or $\frac{d}{dx}F_2(x) > \frac{d}{dx}F_2(x+2) > \frac{d}{dx}F_2(x+4) > \cdots > \frac{d}{dx}F_2(x+2n)$ for $x > 0$; $n \in \mathbb{N}$. But using the asymptotic expansion [1]

$$\psi(x) \sim \ln x + \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} - \frac{1}{252x^6} + \frac{1}{240x^8} + \ldots, \quad x \to \infty \tag{16}$$

we have

$$\lim_{x\to\infty}\frac{d}{dx}F_2(x) = \lim_{x\to\infty}\left\{\frac{31}{4608x^6} - \frac{911}{110592x^8} + \mathcal{O}(x^{-10})\right\} = 0.$$

Then $\frac{d}{dx}F_2(x) > 0$ or the function $F_2(x)$ is increasing for $x > 0$. Also, using the asymptotic expansion (13), we have

$$\lim_{x\to\infty}F_2(x) = \lim_{x\to\infty}\left\{-\frac{31}{23040x^5} + \frac{911}{774144x^7} + \mathcal{O}(x^{-9})\right\} = 0.$$

Hence the function $F_2(x)$ is negative function for $x > 0$, which completes the proof. □

From the results in (10) and (14), we can deduce the following approximation formula.

Theorem 2.1. *The Wallis ratio satisfies the analytic approximation formula*

$$W(x)=\frac{1}{\sqrt{\pi}}\frac{\Gamma\left(x+\frac{1}{2}\right)}{\Gamma(x+1)}=\frac{1}{\sqrt{\pi x}}e^{-\frac{1}{x}\left[8+\frac{1/3}{x^2+\mu(x)}\right]^{-1}},\quad 0\leq\mu(x)\leq\frac{31}{120};\quad x>0 \tag{17}$$

where $\mu(x)$ is continuous function.

Lemma 2.3. *The function*

$$G_3(x)=\frac{31-2304x^5}{4608x^6}+\frac{3\left(24x^2-1\right)}{\left(24x^2+1\right)^2}+\psi(x+1)-\psi\left(x+1/2\right)$$

is completely monotonic for $x>0$. As a consequence, the Wallis ratio satisfies the inequality

$$\frac{\Gamma\left(x+1/2\right)}{\Gamma(x+1)}>\frac{1}{\sqrt{x}}\,e^{-\frac{3x}{24x^2+1}-\frac{31}{23040x^5}},\quad x>0. \tag{18}$$

Proof. Consider the function

$$F_3(x)=-\ln\left(\frac{\sqrt{x}\Gamma\left(x+\frac{1}{2}\right)}{\Gamma(x+1)}\right)-\frac{3x}{24x^2+1}-\frac{31}{23040x^5},\quad x>0$$

and use the relation (11) to get

$$\begin{aligned}\frac{d}{dx}F_3(x)&=G_3(x)\\&=\int_0^\infty\frac{31t^5+e^{t/2}\left(31t^5-276480\right)+69120\left(e^{t/2}+1\right)t\cos\left(\frac{t}{2\sqrt{6}}\right)+276480}{552960\left(e^{t/2}+1\right)}\\&\quad\times e^{-xt}dt.\end{aligned}$$

Using the inequality (12), we have

$$\begin{aligned}&31t^5+e^{t/2}\left(31t^5-276480\right)+69120\left(e^{t/2}+1\right)t\cos\left(\frac{t}{2\sqrt{6}}\right)+276480\\&>\frac{1}{t^2+288}[31e^{t/2}t^7+31t^7+8928e^{t/2}t^5+8928t^5-345600e^{t/2}t^3-345600t^3\\&\qquad-276480e^{t/2}t^2+276480t^2+19906560e^{t/2}t+19906560t-79626240e^{t/2}\\&\qquad+79626240]\\&>\frac{1}{t^2+288}\left[\frac{3602t^7}{7}+\right.\\&\left.\sum_{n=0}^{\infty}\frac{(n+2)(n+4)\left(31n^5+899n^4+12431n^3+109957n^2+582714n+1296720\right)t^{n+8}}{2^{n+1}(n+8)!}\right]\\&>0,\quad t>0.\end{aligned}$$

Hence $G_3(x)$ is completely monotonic function and then $F_3(x)$ is increasing function for $x>0$. Using the asymptotic expansion (13), we get

$$\lim_{x\to\infty} F_3(x) = \lim_{x\to\infty}\left[-\frac{911}{774144x^7} + \mathcal{O}(x^{-9})\right] = 0.$$

Hence the function $F_3(x)$ is negative function for $x > 0$, which completes the proof. □

From the results in (14) and (18), we can deduce the following approximation formula.

Theorem 2.2. *The Wallis ratio satisfies the analytic approximation formula*

$$W(x) = \frac{1}{\sqrt{\pi}}\frac{\Gamma\left(x+\frac{1}{2}\right)}{\Gamma(x+1)} = \frac{1}{\sqrt{\pi x}} e^{-\frac{3x}{24x^2+1}+\frac{31\theta(x)}{23040x^5}}, \quad -1 \le \theta(x) \le 0; \quad x > 0 \tag{19}$$

where $\theta(x)$ is continuous function.

Lemma 2.4. *The function*

$$G_4(x) = \frac{257}{51200x^8} + \frac{5}{20x^2+6} + \frac{1}{2x} - \frac{\left(10x^2+1\right)\left(120x^2+31\right)}{32x^2\left(10x^2+3\right)^2} + \psi(x+1/2) - \psi(1+x)$$

is completely monotonic for $x > 0$. As a consequence, the Wallis ratio satisfies the inequality

$$\frac{\Gamma\left(x+1/2\right)}{\Gamma(x+1)} < \frac{1}{\sqrt{x}}\, e^{-\frac{120x^2+31}{96x\left(10x^2+3\right)}+\frac{257}{358400x^7}}, \quad x > 0. \tag{20}$$

Proof. Consider the function

$$F_4(x) = \ln\left(\frac{\sqrt{x}\Gamma\left(x+\frac{1}{2}\right)}{\Gamma(x+1)}\right) + \frac{120x^2+31}{96x\left(10x^2+3\right)} - \frac{257}{358400x^7}, \quad x > 0$$

and use the relation (11) to get

$$\frac{d}{dx}F_4(x) = G_4(x) = \int_0^\infty \frac{g(t)}{258048000\left(e^{t/2}+1\right)} e^{-xt}dt,$$

where

$$\begin{aligned} g(t) \; &= 257t^7 + e^{t/2}\left(257t^7 - 27776000t + 129024000\right) - 27776000t \\ &\quad -4480000\left(e^{t/2}+1\right)t\cos\left(\sqrt{\frac{3}{10}}t\right) - 129024000, \quad t > 0. \end{aligned}$$

Using the inequality (12), we have

$$\begin{aligned} g(t) &> \frac{1}{t^2+40}(257e^{t/2}t^9+257t^9+10280e^{t/2}t^7+10280t^7-5376000e^{t/2}t^3 \\ &\quad -5376000t^3+129024000e^{t/2}t^2-129024000t^2-1290240000e^{t/2}t \\ &\quad -1290240000t+5160960000e^{t/2}-5160960000) \\ &> \frac{1}{t^2+40}\left[20160t^7+5040t^8+16066t^9/9+\right. \\ &\quad \left.\sum_{n=0}^{\infty}\frac{\left(257(n+7)(n+8)(n+9)(n+10)\left(n^2+5n+16\right)-84000\right)\left(2^{-n-1}t^{n+10}\right)}{(n+3)!(n+7)(n+8)(n+9)(n+10)}\right] \\ &\quad > 0, t>0. \end{aligned}$$

Hence $G_4(x)$ is completely monotonic function and then $F_4(x)$ is increasing function for $x>0$. Using the asymptotic expansion (13), we get

$$\lim_{x\to\infty} F_4(x) = \lim_{x\to\infty}\left[-\frac{3551}{2304000x^9}+\mathcal{O}(x^{-11})\right]=0.$$

Hence the function $F_4(x)$ is negative function for $x>0$, which completes the proof. □

From the results in (10) and (20), we can deduce the following approximation formula.

Theorem 2.3. *The Wallis ratio satisfies the analytic approximation formula*

$$W(x)=\frac{1}{\sqrt{\pi}}\frac{\Gamma\left(x+\frac{1}{2}\right)}{\Gamma(x+1)}=\frac{1}{\sqrt{\pi x}}e^{-\frac{120x^2+31}{96x\left(10x^2+3\right)}+\frac{257\chi(x)}{358400x^7}},\quad 0\le\chi(x)\le 1;\quad x>0 \tag{21}$$

where $\chi(x)$ is continuous function.

3. Comparisons

Remark 1. For the Wallis ratio, the lower bound in inequality (10) is better than the lower bound in inequality (18). Moreover, the upper bound in inequality (20) is better than the upper bound in inequality (14) in a neighborhood of zero, namely for $0<x\lesssim 0.88$ and this behavior is reversed for $x\gtrsim 0.88$.

Remark 2. For the Wallis ratio, the two approximations (7) and (19), for $\chi(x)=1$, have the same speed of convergence since

$$W(x) = \frac{1}{\sqrt{x}}e^{\left(\frac{33}{28672x^7}-\frac{3}{2560x^5}+\frac{\tan^{-1}(2\sqrt{2}x)-\frac{\pi}{2}}{2\sqrt{2}}\right)}\left(1-\frac{55}{32768x^9}+\mathcal{O}(x^{-11})\right),$$
$$x\to\infty$$

and

$$W(x) = \frac{1}{\sqrt{x}} e^{\frac{257}{358400x^7} - \frac{120x^2+31}{96x(10x^2+3)}} \left(1 - \frac{3551}{2304000x^9} + \mathcal{O}(x^{-11})\right), \quad x \to \infty.$$

Remark 3. Now, we will compare the bounds established in Cristea's inequality (9) with the new bounds obtained in (10) and (14). For this purpose, consider the functions

$$F_1(x) = -\frac{3}{2560x^5} - \frac{\cot^{-1}\left(2\sqrt{2}x\right)}{2\sqrt{2}} + \frac{120x^2+31}{96x\left(10x^2+3\right)}, \quad x > 0$$

and

$$F_2(x) = \left(\frac{257}{358400x^7} - \frac{120x^2+31}{96x\left(10x^2+3\right)}\right) - \left(\frac{33}{28672x^7} - \frac{3}{2560x^5} - \frac{\cot^{-1}\left(2\sqrt{2}x\right)}{2\sqrt{2}}\right), \quad x > 0.$$

It is straightforward to verify that both $F_1(x)$ and $F_2(x)$ are increasing functions. Moreover, they satisfy

$$\lim_{x\to\infty} F_1(x) = \lim_{x\to\infty} F_2(x) = 0.$$

Therefore, the lower bound presented in (10) is superior to the corresponding lower bound in Cristea's inequality (9) for all $x > 0$. Likewise, the upper bound given in (14) is superior to the upper bound in (9) for all $x > 0$.

4. Open Question

The functions $\mu(x)$, $\theta(x)$, and $\chi(x)$ have been shown to be continuous and bounded. We now investigate their monotonicity properties. This leads us to the following open question.

Do the functions

$$\mu(x) = -x^2 - \left\{\frac{3}{x}\left[\ln\frac{\sqrt{x}\Gamma\left(x+\frac{1}{2}\right)}{\Gamma(x+1)}\right]^{-1} + 24\right\}^{-1}$$

$$\theta(x) = \frac{23040x^5}{31}\left\{\frac{3x}{24x^2+1} + \ln\frac{\sqrt{x}\Gamma\left(x+1/2\right)}{\Gamma(x+1)}\right\}$$

$$\chi(x) = \frac{358400x^7}{257}\left\{\frac{120x^2+31}{96x\left(10x^2+3\right)} + \ln\frac{\sqrt{x}\Gamma\left(x+1/2\right)}{\Gamma(x+1)}\right\}$$

satisfy the property that $\mu(x)$ and $\chi(x)$ are strictly increasing, while $\theta(x)$ is strictly decreasing for all $x > 0$?

5. Conclusion

This work has examined several completely monotonic functions constructed from the Psi function and explored their analytic implications. Building on these properties, we introduced three new approximation formulas for a Wallis-type ratio in (17), (19) and (21), each accompanied by bounded remainder terms. As a consequence, a collection of new inequalities was obtained, providing refined estimates compared with earlier results. Our analysis also shows that the newly established bounds outperform previously some recent inequalities throughout the positive real axis. To conclude, we raised an open question concerning the monotonicity of the remainder functions associated with the new approximations. Addressing this question may lead to a deeper understanding of the fine behavior of these ratios and could yield further improvements in related inequalities and approximation formulas.

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